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BASE STATION AND CLEANING ROBOT SYSTEM

Abstract

The present disclosure relates to the field of smart home technologies, and proposes a base station and a cleaning robot system. The base station is configured to clean a cleaning system of a cleaning robot, and includes a base station body and a cleaning component. The base station body includes a cleaning groove, and the cleaning component is movably disposed on the base station body. The cleaning component includes a liquid outlet device, and cleaning liquid discharged by the liquid outlet device is configured to clean the cleaning system and enter the cleaning groove. After the cleaning component is positioned opposite to the cleaning system, the cleaning component moves, and the liquid outlet device moves along with the cleaning component. Then, the cleaning component can remove debris from the cleaning system. That is, the cleaning robot can realize automatic cleaning on the cleaning component.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) [0001] The present application is a Continuation Application of U.S. application Ser. No. 18/413,847, which is a Continuation Application of PCT International Application No. PCT/CN2021/136173 filed on Dec. 7, 2021, which claims the priority to Chinese Patent Application No. 202110805998.2 filed on Jul. 16, 2021. The entire contents of the foregoing applications are incorporated herein by reference for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of smart home technologies, and in particular to a base station and a cleaning robot system.

BACKGROUND

[0003] For a conventional cleaning robot, a cleaning part of the cleaning robot typically requires to be cleaned after the cleaning robot performs a wet cleaning task. At present, the wet cleaning part is mostly manually cleaned or replaced with a new one, which is inconvenient to use.

SUMMARY

[0004] According to an aspect of the present disclosure, there is provided a base station for cleaning a cleaning system of a cleaning robot, including: [0005] a base station body, including a cleaning groove; [0006] a cleaning component, movably disposed on the base station body, and including a liquid outlet device, wherein cleaning liquid discharged by the liquid outlet device is configured to clean the cleaning system and enter the cleaning groove.

[0007] According an aspect of the present disclosure, there is provided a cleaning robot system, including the above-mentioned base station and a cleaning robot.

[0008] According to an aspect of the present disclosure, there is provided a base station for cleaning a cleaning system of a cleaning robot, including: [0009] a base station body; [0010] a cleaning component, movably disposed on the base station body, and including a first cleaning part and a second cleaning part different from the first cleaning part; [0011] wherein, the first cleaning part and the second cleaning part remove debris from the cleaning system by interacting with the cleaning system.

[0012] According to an aspect of the present disclosure, there is provided a cleaning robot system, including the above-mentioned base station and a cleaning robot.

[0013] According to an aspect of the present disclosure, there is provided a base station for cleaning a cleaning system of a cleaning robot, including: [0014] a base station body, including a cleaning groove; [0015] a cleaning component, disposed on the base station body approximately horizontally, and including a liquid outlet device and a cleaning part; [0016] wherein, cleaning liquid discharged by the liquid outlet device is configured to clean the cleaning system by the cleaning part and enter the cleaning groove.

[0017] According to an aspect of the present disclosure, there is provided a cleaning robot system, including the above-mentioned base station and a cleaning robot.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] By considering the following detailed description of optional embodiments of the present disclosure in conjunction with the accompanying drawings, various objectives, features, and advantages of the present disclosure will become more apparent. The drawings are merely exemplary illustrations of the present disclosure, and are not necessarily drawn to scale. In the drawings, the same reference numerals always refer to the same or similar elements.

[0019] FIG. **1** is a schematic structural diagram illustrating a cleaning robot of a cleaning robot system in a first posture according to an exemplary embodiment;

[0020] FIG. **2** is a schematic structural diagram illustrating a cleaning robot of a cleaning robot system in a second posture according to an exemplary embodiment;

[0021] FIG. **3** is a schematic structural diagram illustrating a cleaning robot from a first perspective of view according to an exemplary embodiment;

[0022] FIG. **4** is a schematic structural diagram illustrating a cleaning robot from a second perspective of view according to an exemplary embodiment;

[0023] FIG. **5** is a partial exploded structural diagram illustrating a cleaning robot according to an exemplary embodiment;

[0024] FIG. **6** is a schematic structural diagram illustrating a cleaning robot from a third perspective of view according to an exemplary embodiment;

[0025] FIG. **7** is a schematic structural diagram illustrating a liquid container of a cleaning robot according to an exemplary embodiment;

[0026] FIG. **8** is a schematic cross-sectional structural diagram illustrating a liquid container of a cleaning robot according to an exemplary embodiment;

[0027] FIG. **9** is a schematic structural diagram illustrating a cleaning robot from a fourth perspective of view according to an exemplary embodiment;

[0028] FIG. **10** is a schematic structural diagram illustrating a supporting plate of a cleaning robot according to an exemplary embodiment;

[0029] FIG. **11** is a schematic structural diagram illustrating a part of a base station according to an exemplary embodiment;

[0030] FIG. **12** is a schematic structural diagram illustrating a part of a base station according to an exemplary embodiment;

[0031] FIG. **13** is a schematic diagram illustrating the inner structure of a base station from a first perspective of view according to an exemplary embodiment;

[0032] FIG. **14** is a schematic structural diagram illustrating a base station from a second perspective of view according to an exemplary embodiment;

[0033] FIG. **15A** is a schematic partial structural diagram illustrating a cleaning component of a base station according to an exemplary embodiment;

[0034] FIG. **15B** is a schematic partial structural diagram illustrating a cleaning component of a base station according to an exemplary embodiment;

[0035] FIG. **15C** is a schematic partial structural diagram illustrating a cleaning component of a base station according to another exemplary embodiment;

[0036] FIG. **15D** is a schematic partial structural diagram illustrating a cleaning component of a base station according to another exemplary embodiment;

[0037] FIG. **16** is a schematic cross-sectional structural diagram illustrating a cleaning component of a base station according to an exemplary embodiment;

[0038] FIG. **17** is a schematic diagram illustrating a separated structure of a liquid container, a liquid filling connector, and a first fixture part of a cleaning robot system according to an exemplary embodiment;

[0039] FIG. **18** is a schematic structural diagram illustrating a liquid filling connector and a first fixture part of a base station according to an exemplary embodiment;

[0040] FIG. **19** is a schematic structural diagram illustrating a base station in one state according to another exemplary embodiment;
[0041] FIG. **20** is a schematic structural diagram illustrating a base station in another state according to another exemplary embodiment;
[0042] FIG. **21** is a schematic diagram illustrating a cooperative structure between a cleaning robot of a cleaning robot system and a base station according to an exemplary embodiment;
[0043] FIG. **22** is a schematic diagram illustrating a partial cooperation structure between a cleaning robot of a cleaning robot system and a base station according to an exemplary embodiment; and
[0044] FIG. **23** is a schematic partial structural diagram illustrating a base station according to an exemplary embodiment.

[0045] The reference numbers are explained as follows: [0046] **10**, cleaning robot; **110**, robot body; **111**, forward portion; **112**, rearward portion; **120**, sensing system; **121**, position determining device; **122**, bumper; **1221**, through hole; **130**, control module; **140**, driving system; **141**, driving wheel module; **142**, driven wheel; **150**, cleaning system; **151**, dry cleaning system; **152**, side brush; **160**, power system; **170**, human-computer interaction system; **400**, wet cleaning system; **410**, cleaning head; **420**, driving unit; **421**, driving plate; **422**, supporting plate; **4217**, liquid outlet device; **4218**, pump pipe; **4219**, pump; **12**, first charging contactor; **13**, liquid container; **14**, second fixture part; **16**, liquid filling port; **17**, valve; **18**, pipeline; **19**, rotating wheel; [0047] **20**, base station body; **21**, cleaning groove; **211**, liquid pumping port; **22**, guide plate; **221**, slip-resistance protrusion; **222**, extension plate; **23**, guide side surface; **231**, side surface; **232**, middle surface; **24**, guide ceiling; **25**, guide block; **26**, guide wheel; **27**, guide bridge; [0048] **30**, cleaning component; **31**, first cleaning part; **311**, first rotating shaft; **32**, second cleaning part; **33**, cleaning component holder; **34**, driving part; **341**, gear; **342**, rack; **343**, second rotating shaft; **35**, liquid outlet; **36**, liquid outlet device; **371**, first gear; **372**, second gear; **373**, third gear; **374**, fourth gear; **375**, fifth gear; **376**, sixth gear; **377**, seventh gear; **378**, eighth gear; **379**, ninth gear; **40**, second charging contactor; **50**, liquid filling connector; **51**, main body part; **52**, sealing part; **53**, joint part; **60**, first fixture part; **61**, accommodation part; **70**, liquid supply part; **71**, collection container.

DETAILED DESCRIPTION

[0049] Embodiments embodying features of the present disclosure will be described in detail in the following description. It should be understood that the present disclosure can have various changes in different embodiments, which do not depart from the scope of the present disclosure. The description and drawings therein are essentially for illustrative purposes, rather than limiting the present disclosure.

[0050] In the following description of different exemplary embodiments of the present disclosure, reference is made to the accompanying drawings, which form a part of the present disclosure, and show by way of example different exemplary structures, systems, and steps that can implement various aspects of the present disclosure therein. It should be understood that other specific solutions of components, structures, exemplary devices, systems, and steps can be used, and structural and functional modifications can be made without departing from the scope of the present disclosure. Moreover, although the terms “above”, “between”, “within”, etc. may be used in this specification to describe different exemplary features and elements of the present disclosure, these terms are used herein for convenience only. For example, the description may be based on an example direction in the drawings. Nothing in this specification should be understood as requiring a specific three-dimensional direction of the structure to fall within the scope of the present disclosure.

[0051] As illustrated in FIGS. **1** to **23**, a cleaning robot system of embodiments of the present disclosure may include a cleaning robot **10** and a base station.

[0052] In an embodiment of the present disclosure, as shown in FIGS. **3** and **4**, a cleaning robot **10** may include a robot body **110**, a sensing system **120**, a control module **130**, a driving system **140**, a

cleaning system **150**, a power system **160**, and a human-computer interaction system **170**.

[0053] As shown in FIG. 3, the robot body **110** may include a forward portion **111** and a rearward portion **112**, which have approximate circular shapes (i.e., the forward portion and the rearward portion are both circular). The robot body **110** may also have other shapes, including but not limited to an approximate D-shape with a rectangular forward portion and a circular rearward portion, and a shape with the forward portion and the rearward portion being rectangular or square.

[0054] As shown in FIG. 3, the sensing system **120** may include a position determining device **121** located on the robot body **110**, a collision sensor and a proximity sensor disposed on a bumper **122** of the forward portion **111** of the robot body **110**, a cliff sensor disposed on a lower part of the robot body, and a sensing device such as a magnetometer, an accelerometer, a gyroscope, an odometer disposed inside the robot body, for providing the control module **130** with various position information and motion state information of the robot. The position determining device **121** includes, but is not limited to, a camera and a laser distance sensor (LDS).

[0055] As shown in FIG. 3, the forward portion **111** of the robot body **110** may carry the bumper **122**. In a case that a driving wheel module **141** pushes the cleaning robot **10** to walk on the ground during a cleaning process, the bumper **122** detects one or more events along a travel path of the cleaning robot **10** through a sensor system disposed thereon, such as an infrared sensor. The cleaning robot **10** may control the driving wheel module **141**, based on the events detected by the bumper **122**, such as obstacles and walls, such that the cleaning robot **10** responds to the events, such as moving away from the obstacles.

[0056] The control module **130** is disposed on a main circuit board in the robot body **110**, and may include a computing processor, such as a central processing unit, an application processor, for communicating with a non-transitory memory, such as a hard disk, a flash memory, a random access memory. The application processor uses a positioning algorithm such as simultaneous localization and mapping (SLAM) to create a simultaneous map of the environment where the cleaning robot **10** is located, according to obstacle information fed back by the laser distance sensor. In addition, in combination with distance information and speed information fed back by the sensing device such as the sensor, the cliff sensor, the magnetometer, the accelerometer, the gyroscopes, the odometer disposed on the bumper **122**, it is determined comprehensively which operating state the cleaning robot **10** is currently in, which position the cleaning robot **10** is currently located at, and the current posture of the cleaning robot **10**, such as crossing a threshold, moving onto the carpet, being on a cliff, getting stuck on top or bottom, with a full dust container, being picked up, etc. The next action strategy may be provided for different situations, so that the cleaning robot **10** has better cleaning performance and user experience.

[0057] As shown in FIG. 4, the driving system **140** may manipulate the robot body **110** to travel across the ground based on a driving command having distance and angle information (for example, x, y, and θ components). The driving system **140** includes a driving wheel module **141** that can control the left wheel and the right wheel at the same time. In order to more accurately control the motion of the robot, the driving wheel module **141** may include a left driving wheel module and a right driving wheel module. The left and right driving wheel modules are disposed along a transverse axis defined by the robot body **110**. In order for the cleaning robot **10** to move more stably on the ground or have stronger motion ability, the cleaning robot **10** may include one or more driven wheels **142**, which include, but are not limited to, universal wheels. The driving wheel module includes a walking wheel, a driving motor, and a control circuit for controlling the driving motor. The driving wheel module may be also connected to a circuit for measuring the driving current and the odometer. The driving wheel module **141** may be detachably connected to the robot body **110**, which is convenient for disassembly, assembly and maintenance. The driving wheel may have a biased drop type of suspension system, which is fastened in a movable way, for example, attached in a rotatable way, to the robot body **110**, and receives a spring bias that is biased downward and away from the robot body **110**. The spring bias allows the driving wheel to maintain

contact and traction with the ground by a certain ground force, and at the same time, the cleaning part of the cleaning robot **10** also contacts the ground by a certain pressure.

[0058] The power system may include a rechargeable battery, such as a nickel-metal hydride battery and a lithium battery. The rechargeable battery may be connected with a charging control circuit, a battery pack charging temperature detection circuit, and a battery undervoltage monitoring circuit. The charging control circuit, the battery pack charging temperature detection circuit, and the battery undervoltage monitoring circuit are then connected with the single-chip control circuit. The robot is connected to a charging station through a charging electrode disposed on a side or below the robot for charging.

[0059] The human-computer interaction system **170** may include a button on a host panel for a user to select functions. It may also include a display screen and/or an indicator light and/or a speaker, wherein the display screen, the indicator light and the speaker show the user the current state of the robot or the function selection options. It may also include a mobile client application. For a path-navigation type of cleaning robot, the mobile client can show the user the map of the environment where the robot is located, and the location of the robot, which may provide the user with richer and more user-friendly functional items.

[0060] The cleaning system may be a dry cleaning system **151** and/or a wet cleaning system **400**.

[0061] As shown in FIG. **4**, the dry cleaning system **151** according to an embodiment of the present disclosure may include a rolling brush, a dust container, a fan, and an air outlet. The rolling brush that has a certain interaction with the ground sweeps the garbage on the ground and rolls it to the front of a dust suction port between the rolling brush and the dust container. Then the garbage is sucked into the dust container by the suction gas generated by the fan and passing through the dust container. The dust removal capacity of the cleaning robot **10** can be characterized by dust pickup efficiency (DPU). The dust pickup efficiency DPU is affected by the structure and material of the rolling brush, and is affected by the dust suction port, the dust container, the fan, the air outlet, and air utilization of the air duct formed by connecting parts between these four components. Further, the dust pickup efficiency DPU is affected by the type and power of the fan. This is a complicated system scheme problem. Compared with an ordinary plug-in vacuum cleaner, the improvement of dust removal capacity is of greater significance to the cleaning robot with limited power. The improvement of dust removal capacity directly and effectively reduces the power requirements. That is to say, an original robot that may clean 80 square meters of ground with a single charge may evolve into a single charge to clean 180 square meters or more. In addition, the service life of the battery that reduces recharge cycles may also be greatly increased, so that the replacement frequency of the battery by the user may also decrease. More intuitively and more importantly, the improvement of dust removal capability is the most obvious and important user experience, and the user will directly draw a conclusion about whether the sweep is clean or the mopping is clean. The dry cleaning module may also include a side brush **152** having a rotating shaft, wherein the rotating shaft is at a certain angle with respect to the ground, for moving scrap to a rolling brush area of the cleaning system **150**.

[0062] As shown in FIGS. **4** to **8**, the wet cleaning system **400** according to an embodiment of the present disclosure may include: a cleaning head **410**, a driving unit **420**, a liquid delivery mechanism, a liquid container **13**, and the like. The cleaning head **410** may be disposed below the liquid container **13**, and the cleaning liquid inside the liquid container **13** is delivered to the cleaning head **410** through the liquid delivery mechanism, so that the cleaning head **410** performs wet cleaning on a surface to be cleaned. In other embodiments of the present disclosure, the cleaning liquid inside the liquid container **13** may also be sprayed directly onto the surface to be cleaned, and the cleaning head **410** performs cleaning on the surface by evenly spreading the cleaning liquid.

[0063] The cleaning head **410** is configured to clean the surface to be cleaned, and the driving unit **420** is configured to drive the cleaning head **410** to substantially reciprocate along the target

surface, which is a part of the surface to be cleaned. The cleaning head **410** reciprocates along the surface to be cleaned, and a side of the contact surface between the cleaning head **410** and the surface to be cleaned is provided with a cleaning cloth or a cleaning plate, which leads to high-frequency friction with respect to the surface to be cleaned due to the reciprocating motion, thereby removing stains from the surface to be cleaned.

[0064] The higher the friction frequency, the more friction times per unit time. The high-frequency reciprocating motion, also called reciprocating vibration, has a greater cleaning capacity than the ordinary reciprocating motion, such as rotation. The friction cleaning, of which the friction frequency is optionally close to the sound wave, have a better cleaning effect than the rotating friction cleaning with dozens of turns per minute. On the other hand, hair tufts on the surface of the cleaning head **410** may be more uniform and stretched in the same direction under the shaking of high-frequency vibration. Thus, the overall cleaning effect is more uniform, which is reflected in the effect that water marks on the surface to be cleaned after high-frequency vibration cleaning are more uniform, and no messy water mark is left. Instead, if the down pressure is applied to increase the friction force for improving the cleaning effect with low frequency rotation, the down pressure does not cause the hair tufts to extend in the same direction.

[0065] The reciprocating motion may be repeated motion along any one or more directions within the surface to be cleaned. Alternatively, it may be vibrations perpendicular to the surface to be cleaned, which is not strictly limited. Optionally, the direction of the reciprocating motion of the cleaning module is approximately perpendicular to the travel direction of the robot, because the direction of the reciprocating motion parallel to the travel direction of the robot would cause instability to the robot itself, due to the thrust and resistance in the travel direction leading to the driving wheels easy to slip. The impact of slipping is more obvious in a case that the wet cleaning module is included, because the wet surface to be cleaned increases the possibility of slipping. In addition to affecting the smooth moving and cleaning effect of the robot, the slipping may also cause the sensor such as the odometer, the gyroscope to be inaccurate in range measurement. This results in the navigation-type of cleaning robot being unable to accurately locate and create the map. In the case of frequent slipping, the impact on SLAM may not be ignored. Therefore, it is necessary to avoid the robot from slipping as much as possible. In addition to slipping, the motion component of the cleaning head **410** in the travel direction of the robot causes the robot to be continuously pushed forwards and rearwards when traveling, which will lead the traveling of the robot to be unstable and unsmooth.

[0066] In an embodiment of the present disclosure, as shown in FIG. 5, the driving unit **420** may further include: a driving plate **421** connected to a bottom surface of the robot body **110** and configured to provide a driving force; and a supporting plate **422** detachably connected to the driving plate **421** and configured to support the cleaning head **410**, wherein the supporting plate **422** may be raised and lowered under the driving of the driving plate **421**.

[0067] In an optional embodiment of the present disclosure, the wet cleaning system **400** may be connected to the robot body **110** through an active lifting module. In a case that the wet cleaning system **400** is temporarily not in operation, for example, the cleaning robot **10** stops at the base station to clean the cleaning head **410** of the wet cleaning system **400** and injects liquid into the liquid container **13**. Alternatively, in a case that the surface to be cleaned cannot be cleaned by the wet cleaning system **400**, the wet cleaning system **400** is lifted by the active lifting module.

[0068] In the wet cleaning system **400** according to an embodiment of the present disclosure, the cleaning head **410**, the driving plate **421**, the supporting plate **422**, the liquid delivery mechanism, and the liquid container **13** may be powered by one motor or multiple motors. The power system **160** provides power for the motor, and the control module **130** provides the overall control.

[0069] The liquid delivery mechanism in an embodiment of the present disclosure may include a liquid outlet device, which may be directly or indirectly connected to the liquid outlet of the liquid container **13**. As shown in FIG. 10, the cleaning liquid may flow to the liquid outlet device **4217**

through the cleaning liquid outlet of the liquid container **13**, and may be evenly sprayed on the surface to be cleaned by the liquid outlet device. The liquid outlet device may be provided with a connecting part, and the liquid outlet device is connected with the cleaning liquid outlet of the liquid container **13** through the connecting part. The liquid outlet device is provided with a distribution opening, which may be a continuous opening or a combination of several discontinuous small openings. Several nozzles may be disposed at the distribution opening. The cleaning liquid flows to the distribution opening through the cleaning liquid outlet of the liquid container **13** and the connecting part of the liquid outlet device, and is evenly sprayed on the surface to be cleaned through the distribution opening.

[0070] As shown in FIGS. **5** and **10**, the liquid delivery mechanism may further include a pump **4219** and/or a pump pipe **4218**. The pump **4219** may directly communicate with the cleaning liquid outlet of the liquid container **13** or communicate with it through the pump pipe **4218**.

[0071] The pump **4219** may be connected to the connecting part of the liquid outlet device, and may be configured to pump the cleaning liquid from the liquid container **13** to the liquid outlet device. The pump can be a gear pump, a vane pump, a plunger pump, a peristaltic pump, and so on.

[0072] The liquid delivery mechanism pumps out the cleaning liquid in the liquid container **13** through the pump **4219** and the pump pipe **4218**, and then transports it to the liquid outlet device. The liquid outlet device **4217** may be a nozzle, drip hole, soaking cloth, etc., and evenly spread the liquid on the cleaning head **410**, so as to wet the cleaning head **410** and the surface to be cleaned. The stains on the wet surface to be cleaned may be cleaned more easily. In the wet cleaning system **400**, the power/flow rate of the pump may be adjusted.

[0073] In an embodiment of the present disclosure, the liquid container **13** may further include a liquid filling port **16**. As shown in FIGS. **7** to **9**, the liquid filling port **16** may be located on a side wall of the liquid container **13**. In a case that the cleaning robot **10** stops at the base station, the base station can fill the liquid container **13** of the cleaning robot **10** with liquid through the liquid filling port **16**.

[0074] In an embodiment of the present disclosure, as shown in FIG. **7**, a second fixture part **14** may be disposed on the liquid container **13**, and is configured to connect with the base station, so that the base station can fill the liquid container **13** of the cleaning robot **10** with liquid through the liquid filling port **16**.

[0075] In an embodiment of the present disclosure, as shown in FIG. **8**, the liquid filling port **16** of the liquid container **13** may be provided with a valve **17**, which may be opened and closed to control the communication and disconnection between the liquid filling port **16** and the liquid container **13**. A pipeline **18** is disposed in the liquid container **13**, and one end of the pipeline **18** is provided with the valve **17**.

[0076] In an embodiment of the present disclosure, the valve **17** may be an electronic valve or a manual valve, and it is ensured that the valve **17** may be open or closed under the respective control. In other embodiments of the present disclosure, the valve **17** may also be a check valve. In a case that the liquid container **13** is refilled and the connection between the liquid filling port **16** and the liquid container **13** is disconnected, the valve **17** is automatically closed to prevent the cleaning fluid in the liquid container **13** from flowing out. For example, the valve **17** may be a cross valve, a lift check valve, a swing check valve, or the like.

[0077] In an embodiment of the present disclosure, the cleaning robot **10** further includes a first charging contactor **12**, which may be disposed on the robot body **110** and connected to the power system of the cleaning robot **10**. In a case that the cleaning robot **10** stays at the base station, the base station may charge the power system of the cleaning robot **10** through the first charging contactor **12**. In an embodiment of the present disclosure, the first charging contactor **12** may be located on a side of the robot body of the cleaning robot **10**. This arrangement can prevent the stagnant liquid on the ground from polluting the first charging contactor **12**, and also prevent damage to the cleaning robot **10** in a case that the charging contactor contacts the liquid if the robot

10 stays at the base station to fill the liquid container **13** with liquid or clean the cleaning system **150** of the cleaning robot **10**.

[0078] In an embodiment of the present disclosure, as shown in FIG. **11**, the base station may include a base station body **20** and a cleaning component **30**. The cleaning component **30** is movably disposed on the base station body **20**. The base station body **20** may include a cleaning groove **21**, and the cleaning component **30** includes a liquid outlet device **36**. The cleaning liquid discharged by the liquid outlet device **36** is configured to clean the cleaning system **150** of the cleaning robot **10** and enter the cleaning groove **21**.

[0079] In an embodiment of the present disclosure, the liquid outlet device **36** of the base station is movably disposed, so that the cleaning liquid may be sprayed or smeared to the cleaning system **150** of the cleaning robot **10** more evenly. Thus, it is ensured that the cleaning fluid soaks the cleaning system **150** in time if the cleaning component **30** cleans the cleaning system **150** of the cleaning robot **10**.

[0080] In addition, in addition to realizing the cleaning of the cleaning system **150**, it is also avoided that the cleaning liquid overflows and flows into the external environment, or flows onto relevant electrical components of the cleaning robot **10**, thereby causing safety problems.

[0081] In an embodiment of the present disclosure, in a case that the cleaning robot **10** stays at the base station body **20** and keeps stationary, the cleaning component **30** is in contact with the cleaning system **150** of the cleaning robot **10** and moves relative to the base station body **20** and the cleaning robot **10**. The cleaning liquid discharged from the liquid outlet device **36** soaks the cleaning system **150** of the cleaning robot **10** and may effectively clean the cleaning system **150** of the cleaning robot **10**.

[0082] It should be noted that if the cleaning component **30** of the base station moves, the cleaning liquid discharged by the liquid outlet device **36** may be configured to clean the cleaning system **150** of the cleaning robot **10**. That is, the debris on the cleaning system **150** of the cleaning robot **10** can be removed with the help of the cleaning liquid, and the cleaning system **150** is evenly soaked during the movement of the liquid outlet device **36**.

[0083] In an embodiment of the present disclosure, as shown in FIG. **11**, the cleaning component **30** may further include a cleaning component holder **33** movably disposed on the base station body **20**, and the liquid outlet device **36** may be disposed on the cleaning component holder **33**. That is to say, the cleaning component holder **33** serves as a moving part to ensure that the liquid outlet device **36** moves with it. Thus, it is ensured that the cleaning liquid is discharged from different positions and evenly wets an object to be cleaned.

[0084] In an embodiment of the present disclosure, the base station may further include a liquid delivery channel. One end of the liquid delivery channel is in communication with a liquid supply part **70**, and the other end of the liquid delivery channel is in communication with the liquid outlet device **36**, so that the liquid supply part **70** supplies the cleaning liquid into the liquid outlet device **36** via the liquid delivery channel. At least part of the liquid delivery channel is movably disposed along with the cleaning component holder **33**. The liquid supply part **70** provides storage of the cleaning liquid, and the liquid delivery channel is a delivery part, so as to move along with the cleaning component holder **33**.

[0085] In an embodiment of the present disclosure, the liquid delivery channel is a liquid delivery pipe, and the liquid delivery pipe is connected to the cleaning component holder **33**. That is, the liquid outlet device **36** is disposed on the cleaning component holder **33**, and two ends of the liquid delivery pipe are respectively in communication with the liquid supply part **70** and the liquid outlet device **36**, so as to achieve liquid supply.

[0086] Optionally, the liquid delivery channel is provided with a pump. The cleaning liquid in the liquid supply part **70** is transported to the liquid outlet device **36** under the action of the pump, which ensures that the cleaning liquid has a certain impact force, thereby improving the cleaning capability. In an embodiment of the present disclosure, the controller disposed on the base station

may control parameters such as liquid outlet frequency, liquid outlet flux, and liquid outlet duration. Besides, the controller may be further connected with the communication device of the base station. When the communication device receives instructions from the cleaning robot **10** or a remote controller, such as a computer terminal or a mobile application, operations of one or more elements on the base station can be controlled.

[0087] In an embodiment of the present disclosure, a plurality of liquid outlets **35** may be disposed on the liquid outlet device **36** at intervals, the cleaning liquid is discharged through the liquid outlet **35**, and liquid may be discharged at multiple positions to improve cleaning efficiency.

[0088] Optionally, the liquid outlet device **36** may be integrated into the cleaning component holder **33**, and the plurality of liquid outlets **35** is disposed on the cleaning component holder **33** at intervals, so as to realize liquid discharging at multiple positions. In other embodiments of the present disclosure, the liquid outlet device **36** may be also disposed separately on the cleaning component holder **33**, so as to facilitate repair, replacement, and so on of the liquid outlet device **36**.

[0089] In an embodiment of the present disclosure, as shown in FIG. **13**, the cleaning groove **21** disposed below the cleaning component **30** may be provided with a liquid pumping port **211**, and the cleaning liquid in the cleaning groove **21** may be discharged through the liquid pumping port **211**, so as to ensure the cleaning liquid in the cleaning groove **21** to be replaced in time.

[0090] In an embodiment of the present disclosure, as shown in FIG. **14**, the base station further includes a collection container **71**, and the collection container **71** is in communication with the cleaning groove **21** through the liquid pumping port **211**, so that the waste liquid in the cleaning groove **21** flows into the collection container **71**.

[0091] Specifically, as shown in FIG. **14**, the base station further includes a liquid supply part **70**, which is in communication with the liquid outlet **35** through the liquid delivery pipe, and is configured to provide the cleaning liquid for cleaning the cleaning system **150** of the cleaning robot **10**.

[0092] In an embodiment of the present disclosure, the base station further includes a first pump and a second pump. The first pump is configured to pump the cleaning liquid into the cleaning groove **21**. The second pump is configured to pump out the cleaning liquid from the cleaning groove **21**. The first pump and the second pump respectively achieve the feeding and the pumping of the cleaning liquid, so as to ensure the replacement of the cleaning liquid in the cleaning groove **21** and ensure the cleaning effect.

[0093] The first pump is in communication with the liquid supply part **70**, so that the cleaning liquid in the liquid supply part **70** is pumped into the cleaning groove **21** through the liquid outlet **35**. The second pump is in communication with the collection container **71**, so that the cleaning liquid in the cleaning groove **21** is pumped into the collection container **71** through the liquid pumping port **211**.

[0094] In an embodiment of the present disclosure, the first pump and the second pump may be in operation at the same time. The first pump sprays the cleaning liquid into the cleaning groove **21**, and the second pump pumps the cleaning liquid out from the cleaning groove **21**. That is, the cleaning liquid flows rapidly in the cleaning groove **21**.

[0095] In an embodiment of the present disclosure, as shown in FIG. **12** and FIG. **15A**, the cleaning component **30** further includes a cleaning part disposed on the cleaning component holder **33**, wherein the cleaning part may be parallel to the liquid outlet device **36**. This arrangement provides the cleaning component **30** with a compact structure, and the compact structure can ensure that the cleaning system **150** of the cleaning robot **10** is soaked in time by the cleaning liquid discharged by the liquid outlet device **36**, if the cleaning part is performing a cleaning operation, thereby helping the cleaning part to clean the cleaning system **150** of the cleaning robot **10**.

[0096] It should be noted that the cleaning part is parallel to the liquid outlet device **36**. That is, an extension direction of the cleaning part is parallel to a straight line formed by connecting center

points of the plurality of liquid outlets **35** of the liquid outlet device **36**.

[0097] In an embodiment of the present disclosure, the cleaning component **30** may be disposed on the base station body **20** approximately horizontally, as shown in FIGS. **11** and **12**. With such arrangement, the cleaning robot **10** may stay at the base station approximately horizontally and perform the cleaning operation, which reduces the risk of the cleaning robot falling out of the charging station due to an excessive tilt. In an embodiment of the present disclosure, the cleaning robot **10** may stay at the base station safely and bring the cleaning component **30** onto the cleaning part without any help from another device. In addition, since the cleaning component **30** is provided with the liquid outlet device **36** and the cleaning part, the approximate horizontal arrangement of the cleaning component **30** helps to better control the cleaning liquid so as to fall into the cleaning groove **21** smoothly, and prevent the cleaning liquid from sliding along the inclined cleaning component **30** toward outside of the groove.

[0098] In an embodiment of the present disclosure, as shown in FIG. **15A**, the cleaning part may include a first cleaning part **31**, the first cleaning part **31** is disposed on the cleaning component holder **33**, and the first cleaning part **31** removes the debris from the cleaning system **150** of the cleaning robot **10** due to contact and movement relative to the cleaning system **150**. In addition, the liquid outlet **35** on the liquid outlet device **36** may be disposed to face the first cleaning part **31**. With this arrangement, the cleaning liquid discharged from the liquid outlet **35** may be sprayed to the first cleaning part **31** first, facilitating the first cleaning part **31** to uniformly spread the cleaning liquid to the cleaning system **150** of the cleaning robot **10**. In other embodiments of the present disclosure, the cleaning liquid discharged from the liquid outlet **35** may also be directly sprayed to the cleaning system **150** of the cleaning robot **10**, which is not limited in the present disclosure. In an embodiment of the present disclosure, the first cleaning part **31** may be a cleaning roller or the like that rotates around an axis parallel to the liquid outlet device **36**, such as a brush roller or a soft rubber roller.

[0099] In an embodiment of the present disclosure, as shown in FIG. **15A**, the cleaning part may further include a second cleaning part **32**, the second cleaning part **32** is disposed on the cleaning component holder **33**, and the second cleaning part **32** removes debris on the cleaning system **150** in cooperation with the first cleaning part **31** due to contact and movement relative to the cleaning system **150**. The second cleaning part **32** may be disposed in parallel on any side of the first cleaning part **31**. If there are multiple second cleaning parts **32**, the second cleaning parts **32** may be distributed in parallel on any side or both sides of the first cleaning part **31**. As shown in FIG. **15A**, the second cleaning part **32** is disposed on a side of the first cleaning part **31** and above the liquid outlet device **36**. In an embodiment of the present disclosure, the second cleaning part **32** may be a soft rubber scraper or the like.

[0100] Specifically, in a case that the cleaning robot **10** moves to the base station body **20**, the cleaning component **30** is positioned opposite to the cleaning system **150**, the cleaning component **30** moves relative to the base station body **20**, and the first cleaning part **31** and/or the second cleaning part **32** may be in contact with the cleaning system **150**, so as to remove debris on the cleaning system **150**. That is, the cleaning robot **10** can perform automatic cleaning on the cleaning component **30**.

[0101] In an embodiment of the present disclosure, as described above, the cleaning system **150** of the cleaning robot **10** may include a dry cleaning system **151** and a wet cleaning system **400**. The following specifically introduces a cleaning process of the wet cleaning system **400** of the cleaning robot **10** by the cleaning component **30** of the base station.

[0102] As shown in FIG. **1**, in a case that the cleaning robot **10** moves to the base station body **20**, the wet cleaning system **400** of the cleaning robot **10** is fixedly disposed relative to the base station body **20**. The cleaning component **30** of the base station is in contact with the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10**. In other embodiments of the present disclosure, the wet cleaning system **400** of the cleaning robot **10** may realize vertical movement

with the help of the active lifting module. Therefore, in a case that the cleaning robot **10** stops at the base station for the cleaning operation, the active lifting module may be adjusted to achieve better contact between the cleaning component **30** of the base station and the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10**. For example, in an embodiment of the present disclosure, the wet cleaning system **400** of the cleaning robot **10** may be cleaned if it is fully lifted. In other embodiments of the present disclosure, the wet cleaning system **400** of the cleaning robot **10** may be also cleaned in other lifting states. The lifting state of the wet cleaning system **400** may be adjusted according to the material of the cleaning head **410** of the wet cleaning system **400**. For example, in a case that the friction coefficient of the cleaning head **410** to be cleaned is small, the contact between the cleaning head **410** and the cleaning component **30** may be made closer. This ensures that if the cleaning component **30** moves relative to the base station body **20**, the friction between the cleaning head **410** and the cleaning component **30** is within a certain range, which facilitates the proceeding of the cleaning operation; and vice versa. In addition, the lifting state of the wet cleaning system **400** may be adjusted according to the degree of contamination of the cleaning head **410** of the wet cleaning system **400**. For example, in a case that the cleaning head **410** to be cleaned is relatively dirty, the contact between the cleaning head **410** and the cleaning component **30** may be made closer. Thus, a greater friction force is generated between the cleaning head **410** and the cleaning component **30** to ensure that the debris on the cleaning head **410** is effectively removed; and vice versa. In an embodiment of the present disclosure, the lifting state of the wet cleaning system **400** may be adjusted by the user according to the actual situation. Alternatively, a sensor may be disposed at a specific location, such as the cleaning head **410** of the wet cleaning system **400**, and the sensor outputs a specific signal to the control module **130** of the cleaning robot **10**. The control module **130** automatically adjusts the lifting state of the wet cleaning system **400** according to a feedback result of the sensor. In other embodiments of the present disclosure, the lifting state of the wet cleaning system **400** may be also adjusted in other ways, which is not limited in the present disclosure.

[0103] In a case that the cleaning robot **10** is fixed on the base station body **20**, and the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10** is in good contact with the cleaning component **30** of the base station, the cleaning component **30** may perform cleaning on the wet cleaning system **400** of the cleaning robot **10**. In an embodiment of the present disclosure, as shown in FIG. **15A**, the cleaning component **30** includes the first cleaning part **31** in the form of roller and the second cleaning part **32** in the form of scraper. During the cleaning process of the wet cleaning system **400** of the cleaning robot **10** by the cleaning component **30**, the liquid outlet device of the cleaning component **30** may be in operation at the same time to spray the cleaning liquid onto the first cleaning part **31**. The first cleaning part **31** evenly spreads the cleaning liquid onto the cleaning head **410** of the wet cleaning system **400** through contact with the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10** and its own rotation. In addition, the first cleaning part **31** may be a brush roller or a soft rubber roller with blades. The cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10** may be composed of a soft cloth made of fiber material or cotton, or a sponge. During the cleaning process, the bristles or blades of the first cleaning part **31** may penetrate deeply into the cleaning head **410** and is fully contact with it, thereby bringing out the debris in the cleaning head **410** of the wet cleaning system **400**. In addition, the first cleaning part **31** may rotate while moving left and right, and its bristles or blades may produce a flapping effect on the cleaning head **410** of the wet cleaning system **400** during rotation. This causes the debris contained within the cleaning head **410** to be shaken out and scraped off by the vibration generated by the flapping effect. At the same time, in cooperation with the operation of the first cleaning part **31**, the scraper of the second cleaning part **32** scrapes the debris that is taken out or shaken out of the cleaning head **410** of the wet cleaning system **400**, and also scrapes the waste liquid on the cleaning head **410**. In other embodiments of the present disclosure, the first cleaning part **31** may move left and right while rotating in different directions.

For example, in a case that the first cleaning part **31** moves to the left relative to the base station body **20**, it may rotate clockwise at the same time. In a case that the first cleaning part **31** moves to the right relative to the base station body **20**, it may rotate counterclockwise at the same time.

[0104] As mentioned above, the wet cleaning system **400** of the cleaning system **150** may reciprocate relative to the base station body **20**. In an embodiment of the present disclosure, during the movement of the cleaning component **30** relative to the base station body **20**, the wet cleaning system **400** of the cleaning robot **10** may be stationary, or the wet cleaning system **400** may also perform the corresponding reciprocating motion to be in cooperation with the motion of the cleaning component **30**. This ensures that the wet cleaning system **400** can be cleaned quickly. For example, in a case that the cleaning component **30** moves to the left relative to the base station body **20**, the wet cleaning system **400** of the cleaning robot **10** may move to the right relative to the base station body **20**, so as to increase the relative motion speed between the cleaning component **30** and the wet cleaning system **400** for improving the cleaning efficiency; and vice versa.

[0105] In an embodiment of the present disclosure, the first cleaning part **31** and the second cleaning part **32** are synchronously and movably disposed. As shown in FIG. 15A, the first cleaning part **31** and the second cleaning part **32** are both disposed on the cleaning component holder **33** of the cleaning component **30**. Thus, the cleaning component holder **33** drives the first cleaning part **31** and the second cleaning part **32** to move synchronously with the same moving direction, and the first cleaning part **31** and the second cleaning part **32** sequentially complete the cleaning of the cleaning system **150**. In other embodiments of the present disclosure, the first cleaning part **31** and the second cleaning part **32** may be respectively disposed on different holders. With such arrangement, by separately controlling the movement of the holders, motions of the first cleaning part **31** and the second cleaning part **32** can be controlled separately, and the asynchronous movements of the first cleaning part **31** and the second cleaning part **32** can be realized. For example, the first cleaning part **31** or the second cleaning part **32** may be operated independently. Alternatively, a time difference may be formed according to the actual situation in a case that the first cleaning part **31** and the second cleaning part **32** are implemented to clean at the same position of the cleaning head **410**. The present application is not limited in this regard.

[0106] As mentioned above, the cleaning component **30** may include one or more first cleaning parts **31** and second cleaning parts **32**. For example, in an embodiment of the present disclosure, the cleaning component **30** may include two first cleaning parts **31** and one second cleaning part **32**, wherein the first cleaning parts **31** are respectively disposed on two sides of the second cleaning part **32**, as shown in FIG. 15C. In an embodiment, during the reciprocating motion of the cleaning component **30**, the first cleaning part **31** may always be in front of the second cleaning part **32**. This arrangement enables the cleaning component **30** to clean firstly the portions to be cleaned of the cleaning head **410** during the cleaning process of the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10**. That is, the bristles or blades of the first cleaning part **31** produce the flapping effect on the cleaning head **410** during the rotation of the first cleaning part **31**, causing the debris contained in the cleaning head **410** to be shaken out and scraped off by the vibration generated by the flapping effect. Subsequently, the scraper of the second cleaning part **32** scrapes off the debris taken out or shaken out of the cleaning head **410** and the waste liquid on the cleaning head **410**, so as to ensure that the cleaning head **410** can be cleaned more thoroughly.

[0107] In some embodiments of the present disclosure, by controlling the height of the liquid level in the cleaning groove **21**, the first cleaning part **31** and the second cleaning part **32** may be partially immersed in the cleaning liquid in the cleaning groove **21**, completely immersed in the cleaning liquid, or not immersed in the cleaning liquid at all.

[0108] In a case that the first cleaning part **31** and the second cleaning part **32** are partially immersed in the cleaning liquid in the cleaning groove **21**, the first cleaning part **31** performs the reciprocating motion while rotating. During the rotation of the first cleaning part **31**, the cleaning liquid in the cleaning groove **21** may be taken out and applied to the cleaning head **410** of the wet

cleaning system **400** of the cleaning robot **10**. Thus, the cleaning head **410** may be cleaned in a case that the liquid outlet device of the base station is not in operation. In addition, during the reciprocating motions of the first cleaning part **31** and the second cleaning part **32**, the impurities thereon may be removed under the washing of the liquid flow.

[0109] In a case that the first cleaning part **31** and the second cleaning part **32** are all immersed in the cleaning liquid in the cleaning groove **21**, that is, the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10** may be immersed in the cleaning liquid in the cleaning groove **21**, the cleaning head **410** may use the cleaning liquid in the cleaning groove **21** to perform cleaning if the liquid outlet device of the base station is not in operation. In addition, during the reciprocating motions of the first cleaning part **31** and the second cleaning part **32**, the impurities thereon may be removed under the washing of the liquid flow.

[0110] In a case that the first cleaning part **31** and the second cleaning part **32** are not immersed in the cleaning liquid in the cleaning groove **21** at all, the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10** completely uses the cleaning liquid sprayed by the liquid outlet device of the base station for cleaning. Thereby, it is ensured that the cleaning head **410** is not secondarily contaminated by the debris in the cleaning groove **21**, which may be applied to the situation where the cleaning head **410** is seriously dirty, and may be also applied to the situation where the cleaning fluid in the cleaning groove **21** has been used for too many times and has not been replaced.

[0111] In an embodiment of the present disclosure, as shown in FIG. **15A** and FIG. **16**, the base station further includes a driving part **34**, connected to the cleaning component holder **33**, and connected to the base station body **20**, so as to drive the cleaning component holder **33** to move relative to the base station body **20**.

[0112] Optionally, as shown in FIGS. **15A** and **15B**, the driving part **34** and the cleaning component holder **33** move synchronously with respect to the base station body **20**. That is, the driving part **34** may include a motor and a gear **341**, wherein the motor drives the gear **341** to rotate. A rack **342** may be disposed on the base station body **20**, so that the gear **341** moves along the extending direction of the rack **342**, and the driving part **34** and the cleaning component holder **33** move synchronously on the base station body **20**. Optionally, racks **342** are disposed on both sides of the cleaning component holder **33**. Accordingly, there may be at least two gears **341**, and the at least two gears **341** mesh with the two racks **342** respectively.

[0113] In addition, as described above, in a case that the cleaning component **30** moves relative to the base station body **20**, the first cleaning part **31** of the cleaning component **30** also rotates by itself. In an embodiment of the present disclosure, one motor may be configured to drive the cleaning component **30** to move relative to the base station body **20** and at the same time to drive the first cleaning part **31** to rotate by itself. Specifically, an output shaft of the motor is connected to the gear **341** and the first cleaning part **31** through a gear transmission component. Thus, in a case that the motor is running, the motor drives the gear **341** and the first cleaning part **31** to rotate at the same time. In this case, the gear **341** moves along the extending direction of the rack, and the first cleaning part **31** rotates by itself. The gear transmission component is configured according to the actual requirements in rotating speed, and there is no limitation here. The gear transmission component includes a gear and a connection shaft. Further, the gear transmission component may also include a conveyor belt or a chain, etc., which is not limited here, as long as the motor can be ensured to simultaneously drive the gear **341** and the first cleaning part **31** to rotate. In some embodiments of the present disclosure, it is not excluded that two motors are configured to drive the movement of the cleaning component **30** relative to the base station body **20** and the rotation movement of the first cleaning part **31** respectively.

[0114] Optionally, the driving part **34** may be fixed to the base station body **20**, and the driving part **34** may be an air cylinder or an oil cylinder. A telescopic rod of the driving part **34** is connected to the cleaning component holder **33**, so that the cleaning component holder **33** is driven to move on

the base station body **20** due to the extension and retraction of the telescopic rod. In other embodiments of the present disclosure, the driving part **34** may also be an electric cylinder, or configured in such a way that the motor and the conveyor belt cooperate with each other, as long as it can drive the cleaning component holder **33** to move, which is not limited in the present disclosure. As mentioned above, the first cleaning part **31** and the second cleaning part **32** in an embodiment of the present disclosure may be located on different holders, so as to realize the asynchronous movements of the two. For this reason, the holders of the first cleaning part **31** and the second cleaning part **32** may be provided with independent driving parts, which is not limited in the present disclosure.

[0115] In an embodiment of the present disclosure, the left and right movement of the cleaning component **30** relative to the base station body **20** and the rotation of the first cleaning part **31** are driven by the same driving part, as shown in FIG. **15D**. In an embodiment, the left and right movement of the cleaning component **30** and the rotation of the first cleaning part **31** are realized by the driving part **34** in cooperation with a multi-stage gear. In an embodiment, the driving part **34** may be a motor, and the cleaning component **30** may also include a gear transmission component. The motor drives the cleaning component holder **33** to move through the gear transmission component while the first cleaning part **31** rotates. That is, the gear **341** and the first cleaning part **31** are driven to rotate synchronously.

[0116] As shown in FIG. **15D**, the gear transmission component includes a first gear **371**, a second gear **372**, a third gear **373**, a fourth gear **374**, a fifth gear **375**, a sixth gear **376**, a seventh gear **377**, an eighth gear **378**, and a ninth gear **379**. The motor is connected to the first gear **371**, the first gear **371** is meshed with the second gear **372**, the second gear **372** is meshed with the third gear **373**, and the second gear **372** is located between the first gear **371** and the third gear **373**. Therefore, in a case that the motor drives the first gear **371** to rotate, the first gear **371** drives the third gear **373** to rotate through the second gear **372**. The fourth gear **374** is connected to the third gear **373**, and the fourth gear **374** and the third gear **373** are coaxially disposed, so that the third gear **373** drives the fourth gear **374** to rotate synchronously. The fourth gear **374** is meshed with the fifth gear **375** to drive the fifth gear **375** to rotate. The sixth gear **376** is connected to the fifth gear **375**, and the sixth gear **376** and the fifth gear **375** are coaxially disposed, so that the fifth gear **375** drives the sixth gear **376** to rotate synchronously. A first rotating shaft **311** connected to the sixth gear **376** and the fifth gear **375** drives the first cleaning part **31** to rotate. The sixth gear **376** is meshed with the seventh gear **377** to drive the seventh gear **377** to rotate. The eighth gear **378** is connected to the seventh gear **377**, and the eighth gear **378** and the seventh gear **377** are coaxially disposed, so that the seventh gear **377** drives the eighth gear **378** to rotate coaxially. The eighth gear **378** is meshed with the ninth gear **379** to drive the ninth gear **379** to rotate. A second rotating shaft **343** connected to the ninth gear **379** drives the gear **341** disposed thereon to rotate, so that the gear **341** moves along the rack **342**.

[0117] In an embodiment, the motor may realize forward rotation and reverse rotation, and accordingly, the cleaning component holder **33** may be driven to move in two opposite directions, while the first cleaning part **31** may be also driven to rotate in two directions (i.e., clockwise rotation and counterclockwise rotation). For example, the motor may drive the cleaning component holder **33** to move to left relative to the base station body **20**, while driving the first cleaning part **31** to rotate clockwise. The motor may also drive the cleaning component holder **33** to move to right relative to the base station body **20**, while driving the first cleaning part **31** to rotate counterclockwise. It should be noted that the types and sizes of the gears are not limited here, and may be selected according to actual needs.

[0118] In an embodiment of the present disclosure, the liquid outlet **35** of the liquid outlet device **36** may face at least one of the first cleaning part **31** and the second cleaning part **32**, and the cleaning liquid discharged from the liquid outlet **35** may impact on at least one of the first cleaning part **31** and the second cleaning part **32**. That is, the liquid outlet **35** not only serves as a channel for

the cleaning liquid to enter the cleaning groove **21**, but the liquid outlet **35** may also make the liquid flow impact on at least one of the first cleaning part **31**, the second cleaning part **32**, and the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10**, in order to achieve cleaning accordingly.

[0119] In an embodiment of the present disclosure, the first cleaning part **31** and the second cleaning part **32** are disposed side by side, and the liquid outlet **35** of the liquid outlet device **36** is located below the second cleaning part **32** and faces the first cleaning part **31**. The liquid outlet **35** sprays the cleaning liquid in the liquid supply part **70** to the first cleaning part **31**, and interacts with the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10** due to rotation of the first cleaning part **31**, so as to apply the cleaning liquid to the cleaning head **410**. In other embodiments of the present disclosure, a liquid outlet direction of the liquid outlet **35** of the liquid outlet device **36** may be directed toward the cleaning head **410**, and the cleaning liquid is sprayed directly to the cleaning head **410**. The impact of the cleaning liquid on the cleaning head **410** is configured to cooperate with the first cleaning part **31** and the second cleaning part **32** so as to realize the cleaning of the cleaning head **410**.

[0120] In addition, in other embodiments of the present disclosure, the liquid outlet device **36** and the cleaning component **30** (that is, the first cleaning part **31** and the second cleaning part **32**) may be separately disposed. With such arrangement, if some parts cannot operate, other parts are not affected. For example, the base station may only use the liquid outlet device **36** to complete the cleaning of the cleaning head **410**. That is, the impact of the cleaning liquid on the cleaning head **410** is used completely to achieve the cleaning of the cleaning head **410**.

[0121] In an embodiment of the present disclosure, a plurality of liquid outlets **35** may be disposed on the liquid outlet device **36**, and the plurality of liquid outlets **35** may be in operation at the same time, or may discharge the cleaning liquid sequentially according to a preset rule. That is, the plurality of liquid outlets **35** does not discharge the cleaning liquid at the same time. For example, different pumps or valves may be configured to control the liquid discharge time and frequency of different liquid outlets **35**. Such arrangement may adapt the base station suitable to different shapes and sizes of the cleaning heads **410**. For example, in a case that the cleaning area of the cleaning head **410** is small, a part of the plurality of liquid outlets **35** may be controlled to operate, so as to avoid the waste of cleaning liquid.

[0122] The above is mainly directed to the cleaning of the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10**. In other embodiments of the present disclosure, the base station may also clean other elements of the cleaning robot **10**, which is not limited in the present disclosure.

[0123] In an embodiment of the present disclosure, as shown in FIG. **11** and FIG. **12**, the base station further includes a liquid filling connector **50** disposed on the base station body **20**. The liquid filling connector **50** is configured to connect with the liquid filling port **16** of the liquid container **13** of the cleaning robot **10**, so as to realize the liquid filling into the liquid container **13** of the cleaning robot **10**.

[0124] In an embodiment of the present disclosure, the cleaning robot **10** may move to and stay at the base station body **20**, as shown in FIG. **1**, to perform subsequent liquid filling.

[0125] In an embodiment, in a case that the cleaning robot **10** moves to the base station body **20**, the liquid filling connector **50** of the base station may be connected to the liquid filling port **16** of the cleaning robot **10**, so that the base station supplies the liquid into the liquid container **13** through the liquid filling connector **50**.

[0126] In an embodiment of the present disclosure, since the cleaning robot **10** may swing slightly from side to side during docking at the base station, in order to align the liquid filling connector **50** of the base station better with the liquid filling port **16** of the liquid container **13** of the cleaning robot **10**, at least part of the liquid filling connector **50** of the base station is movably disposed. For example, the liquid filling connector **50** is made of a flexible material, or the liquid filling

connector **50** is disposed on the flexible material.

[0127] In an embodiment of the present disclosure, as shown in FIGS. **17** and **18**, the liquid filling connector **50** includes: a body part **51** connected to the base station body **20**; a sealing part **52**, one end of the sealing part **52** being connected to the body part **51**; and a joint part **53**, connected to the other end of the sealing part **52** away from the body part **51**. The joint part **53** is connected to the liquid container **13**, and the sealing part **52** is made of a flexible material.

[0128] Specifically, the body part **51** is a main flow path of the liquid, the joint part **53** is a hard interface part for connecting with the liquid filling port **16** of the liquid container **13** of the cleaning robot **10**, and the sealing part **52** is a soft structure. By disposing the soft sealing part **52**, the liquid filling connector **50** may move radially and axially, which facilitates alignment with the liquid filling port **16** of the liquid container **13**.

[0129] In an embodiment of the present disclosure, as shown in FIG. **17**, the liquid filling port **16** is configured to match with the liquid filling connector **50**. That is, one end of the liquid filling connector **50** may be inserted into the liquid filling port **16**, and further, the joint part **53** of the liquid filling connector **50** is inserted into the liquid filling port **16**. As mentioned above, a valve is disposed at the liquid filling port **16** of the cleaning robot **10**. For example, a cross valve is disposed at the liquid filling port **16**. In a case that the liquid filling connector **50** of the base station is aligned with the liquid filling port **16** of the cleaning robot **10**, the base station starts to fill liquid into the liquid container **13** through the liquid filling port **16**. The cross valve is opened under the action of the liquid pressure from the liquid filling connector **50**, so that the liquid filling port **16** is in communication with the liquid container **13**, and the cleaning liquid flows into the liquid container **13**. In a case that the liquid filling is completed, the liquid pressure at the liquid filling port **16** from the liquid filling connector **50** disappears, and the cross valve is closed. Thus, the liquid filling port **16** is disconnected with the liquid container **13**, thereby avoiding the cleaning liquid in the liquid container **13** from overflowing.

[0130] In an embodiment of the present disclosure, it is also possible to apply a forward driving force onto the driving wheel of the cleaning robot **10**, if the cleaning robot **10** stays at the base station to fill liquid into the liquid container **13**. During the process of filling liquid into the liquid container **13**, the liquid filling connector **50** of the base station generates a backward thrust on the cleaning robot **10** while discharging liquid, rendering the cleaning robot **10** to tend to move backward. Increasing the forward driving force on the driving wheel may offset at least a part of the thrust, thereby ensuring that the cleaning robot **10** is more stable if filling the liquid container **13** with liquid. In other embodiments of the present disclosure, it may be determined, by the liquid discharge speed of the liquid filling connector **50**, the mass of the cleaning robot **10** itself, or the friction between the driving wheel and the staying surface of the base station if the cleaning robot **10** stays at the base station, whether the forward driving force is increased or not, and what the magnitude of the driving force is, which are not limited in the present disclosure.

[0131] In order to enable the liquid container **13** of the cleaning robot **10** to be filled with cleaning liquid in time, a sensor may be disposed on the cleaning robot **10** to detect the change in liquid level in the liquid container **13**. For example, a float containing a magnetic element may be disposed in the liquid container **13**, and one or more magnetic induction elements may be disposed on the liquid container **13** or the robot body of the cleaning robot **10**, so as to detect the change in liquid level in the liquid container **13**. In a case that the liquid level in the liquid container **13** is lower than a predetermined threshold, the cleaning robot **10** may automatically return to the base station to fill liquid, or the cleaning robot **10** may alert the user by an app or voice, and the user controls the cleaning robot **10** to return to the base station for liquid filling. In other embodiments of the present disclosure, the change in liquid level in the liquid container **13** may also be detected by other means, such as an infrared sensor. In other embodiments of the present disclosure, other control methods may be also configured to help the cleaning robot **10** returning to the base station for liquid filling. For example, the cleaning robot **10** may automatically return to the base station to

fill liquid after completing the task for cleaning a designated cleaning area or the task for cleaning a designated region. The present disclosure is not limited in this respect. In addition, in combination with the foregoing, the liquid container **13** may be filled with liquid while the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10** is cleaned.

[0132] In an embodiment of the present disclosure, as shown in FIGS. **17** and **18**, the base station further includes a first fixture part **60**, disposed on the base station body **20** and used for connection with a second fixture part **14** on the liquid container **13**.

[0133] Specifically, in a case that the cleaning robot **10** moves to the base station body **20**, and the first fixture part **60** is connected to the second fixture part **14**, the liquid filling connector **50** is connected to the liquid container **13**. At this time, the liquid container **13** is filled with liquid through the liquid filling connector **50**.

[0134] In an embodiment of the present disclosure, as shown in FIG. **18**, the first fixture part **60** is formed with an accommodation part **61**, and one end of the liquid filling connector **50** for connecting with the liquid container **13** is located within the accommodation part **61**. In a case that the first fixture part **60** is connected to the second fixture part **14**, the liquid filling connector **50** located within the accommodation part **61** may be reliably connected with the liquid filling port **16** of the liquid container **13**.

[0135] In an embodiment of the present disclosure, as shown in FIG. **17**, the second fixture part **14** is a groove, and the groove is configured to match with the first fixture part **60**. That is, the first fixture part **60** is inserted into the groove, so that the liquid filling connector **50** and the liquid filling port **16** are reliably connected. An outer surface of the first fixture part **60** may be a sloped surface to facilitate insertion into the groove. In the case where the cleaning robot **10** and the base station body **20** are not completely aligned, the first fixture part **60** may also be introduced into the second fixture part **14**.

[0136] In an embodiment of the present disclosure, the liquid supply part **70** may be in communication with the liquid filling connector **50**, and the liquid supply part **70** supplies liquid into the liquid container **13** through the liquid filling connector **50**. The liquid supply part **70** is configured to contain the cleaning liquid, and the liquid in the liquid supply part **70** may be fed into the liquid container **13** through the liquid filling connector **50**.

[0137] Optionally, the liquid supply part **70** may be selectively in communication with the liquid filling connector **50** or the liquid outlet **35**. That is, the liquid supply part **70** may fill liquid into the liquid container **13** through the liquid filling connector **50**, or the liquid supply part **70** may supply the cleaning liquid into the cleaning groove **21** through the liquid outlet **35** of the liquid outlet device **36**. The first pump is configured to pump the cleaning liquid into the cleaning groove **21**, or the first pump is configured to pump the liquid into the liquid filling connector **50**, so as to fill the liquid container **13** with liquid.

[0138] It should be noted that the liquid discharged from the liquid supply part **70** may be shunted into two channels, wherein one channel is in communication with the liquid filling connector **50**, and the other channel is in communication with the liquid outlet **35**. The liquid supply part **70** may be selectively in communication with two channels, so as to control the liquid filling to the liquid filling connector **50** or the liquid outlet **35**. The two channels may be respectively provided with valves, and opening and closing of the valves are controlled to control the disconnection and the communication between the two channels. Alternatively, a three-way valve may be disposed. That is, a three-way valve is configured to control the liquid supply part **70** to be in communication with the respective channel.

[0139] As shown in FIG. **11**, the base station further includes a second charging contactor **40**, and the second charging contactor **40** is configured to electrically connect with the first charging contactor **12** of the cleaning robot **10**, so that the base station may charge the cleaning robot **10**. As shown in FIG. **2**, in a case that the cleaning robot **10** stays at the base station, the second charging contactor **40** is electrically connected to the first charging contactor **12**.

[0140] In some embodiments, as shown in FIG. 12, the base station body 20 further includes a guide side surface 23, the second charging contactor 40 is disposed on the guide side surface 23, and the first charging contactor 12 is disposed on a side surface of the cleaning robot 10. Thus, the second charging contactor 40 may be electrically connected to the first charging contactor 12.

[0141] In some embodiments, as shown in FIG. 12, the guide side surface 23 includes two opposite side surfaces 231 and an middle surface 232 located between the two side surfaces 231, wherein the middle surface 232 is opposite to a forward direction of the cleaning robot 10 moving toward the base station. The second charging contactor 40 is disposed on the middle surface 232. That is, the first charging contactor 12 is disposed on an end side surface of the cleaning robot 10.

[0142] In an embodiment of the present disclosure, the plurality of second charging contactors 40 and the plurality of first charging contactors 12 are all arranged in pairs. Optionally, the second charging contactors 40 may also be located on the side surface 231. That is, the two second charging contactors 40 in a pair may be located on the two side surfaces 231, respectively.

[0143] Accordingly, in an embodiment of the present disclosure, the first charging contactor 12 located on the cleaning robot 10 may be located on the front side of the cleaning robot 10. As shown in FIG. 21, the forward portion of the cleaning robot 10 is provided with a bumper 122, which is movably disposed on the robot body of the cleaning robot 10. In a case that the cleaning robot 10 encounters an obstacle in front of it in motion, the bumper 122 will collide with the obstacle and move toward the robot body of the cleaning robot 10. After the cleaning robot 10 is free from the obstacle, the bumper 122 moves away from the robot body of the cleaning robot 10. Therefore, during the operation of the cleaning robot 10, the bumper 122 may be continuously compressed and extended. In an embodiment of the present disclosure, the first charging contactor 12 of the cleaning robot 10 is disposed on the robot body of the cleaning robot 10 behind the bumper 122, and a through hole 1221 is formed in the corresponding part of the bumper 122. Thus, the first charging contactor 12 is in contact with the second charging contactor 40 during the charging process of the cleaning robot 10. Since the first charging contactor 12 is disposed at the rear part of the bumper 122, it may be prevented from being directly exposed to the outside of the robot body, thereby avoiding the friction damage to the first charging contactor 12 if the cleaning robot 10 collides with a hard obstacle.

[0144] In an embodiment of the present disclosure, the first charging contactor 12 and the wet cleaning system 400 of the cleaning robot 10 are respectively located on opposite sides of the cleaning robot 10, that is, at the front and rear ends in a travelling direction of the cleaning robot 10. Specifically, the first charging contactor 12 is located on the front side of the cleaning robot 10, and the wet cleaning system 400 is located on the rear side of the cleaning robot 10. Therefore, in an embodiment of the present disclosure, the cleaning robot 10 may stay at the base station in two postures. In a case that the cleaning robot 10 returns to the base station for charging, the cleaning robot 10 is travelling forward to stay at the base station. In a case that the cleaning robot 10 cleans the wet cleaning system 400 or fills the liquid container 13, the cleaning robot 10 is travelling in a reverse direction to stay at the base station. In order to cooperate with these two travelling modes, elements for communicating with the base station may be disposed in the front and rear direction of the cleaning robot 10. For example, an infrared device for receiving signals from the base station, etc., is disposed in the front and rear direction of the cleaning robot 10, which is not limited by the present disclosure.

[0145] In an embodiment of the present disclosure, as shown in FIG. 19, the base station may further include a guide bridge 27, disposed above the cleaning groove 21 and configured to support the driven wheel 142 of the cleaning robot 10. As shown in FIG. 4, the driven wheel 142 is disposed in front of the bottom part of the cleaning robot 10. In a case that the cleaning robot 10 stays at the base station for charging, in order to maintain the stability of the cleaning robot 10, a support (that is, the guide bridge 27) may be disposed under the driven wheel 142. As shown in FIG. 19, the guide bridge 27 in an embodiment spans the front and rear ends of the cleaning groove

21, so as to guide the movement of the driven wheel **142** and play a supporting role after the cleaning robot **10** is parked. In other embodiments of the present disclosure, a forwardly extending broken bridge may be disposed only at the front end of the groove, and its extension length may be determined according to factors such as the parking position of the cleaning robot **10** and the disposing position of the driven wheel **142**, which is not limited in the present disclosure. In an embodiment of the present disclosure, since the cleaning component **30** capable of reciprocating left and right is disposed above the cleaning groove, the guide bridge **27** may be movably disposed above the cleaning groove **21**, so as to prevent the guide bridge **27** from obstructing the movement of the cleaning component **30**. For example, in a case that the cleaning robot **10** stays at the base station for charging, the guide bridge **27** may be moved to the middle part of the cleaning groove **21**, so as to guide and support the driven wheel **142** of the cleaning robot **10**. In a case that the cleaning robot **10** stays at the base station to clean the cleaning head **410** of the wet cleaning system **400**, the guide bridge **27** may be moved to one side of the cleaning groove **21**, so that the cleaning component **30** may move left and right. In an embodiment of the present disclosure, as shown in FIG. **19**, the guide bridge **27** and the cleaning component **30** may be disposed on the same holder, and are driven by the same driving part to move to left and right. This arrangement helps to provide a more compact arrangement for various components, and effectively utilize the space of the base station.

[0146] In an embodiment of the present disclosure, as shown in FIG. **19**, the base station body **20** further includes a guide plate **22**, and the guide plate **22** is provided with a slip-resistance protrusion **221**. The cleaning robot **10** moves along the slip-resistance protrusion **221** onto the guide plate **22**. The slip-resistance protrusion **221** may generate a certain friction with respect to the cleaning robot **10**, ensuring that the cleaning robot **10** can reliably move onto the base station body **20**, and can assist the cleaning robot **10** in fixture during the cleaning process.

[0147] In an embodiment of the present disclosure, the cleaning component **30** is located above the guide plate **22**, and the cleaning component **30** and the slip-resistance protrusion **221** are spaced apart. Thus, after the cleaning robot **10** moves a certain distance on the guide plate **22**, the cleaning component **30** is disposed opposite to the cleaning system **150**, so as to perform the subsequent cleaning process.

[0148] Optionally, the cleaning groove **21** is disposed on the guide plate **22**, and the guide plate **22** includes an inclined surface and a flat surface. The slip-resistance protrusion **221** may be disposed on the inclined surface, and the cleaning groove **21** may be disposed on the flat surface.

[0149] It should be noted that the slip-resistance structure formed by the slip-resistance protrusions **221** corresponds to the walking wheel assembly of the cleaning robot **10**. In a case that there are two traveling wheel assemblies, there are also two slip-resistance structures.

[0150] In an embodiment of the present disclosure, as shown in FIGS. **19** and **20**, an extension plate **222** is disposed on the base station body **20**, and the extension plate **222** is connected to an end of the base station body **20**, thereby assisting the cleaning robot **10** in moving to the base station body **20**. The extension plate **222** is foldably disposed. That is, it can be stacked on the guide plate **22**. In a special situation, such as in the case where the floor is slippery, the extension plate **222** may be released to facilitate the cleaning robot **10** to climb.

[0151] In an embodiment of the present disclosure, the base station body **20** further includes a guide ceiling **24** on which a guide part for contacting the cleaning robot **10** is disposed. The guide part is located above the cleaning component **30**. The guide part may limit the cleaning robot **10** in position and ensure that the cleaning robot **10** moves to a suitable position.

[0152] Specifically, the guide part is located above the cleaning component **30**. That is, the cleaning component **30** is located on the guide plate **22**, and the guide part is located on the guide ceiling **24**. Along a height direction, the guide part is located above the cleaning component **30**.

[0153] In an embodiment of the present disclosure, as shown in FIGS. **21** and **22**, the guide part may include a guide block **25**, and a rotating wheel **19** is disposed on the upper edge of the cleaning

robot **10**. As shown in FIG. **9**, the rotating wheel **19** may rotate along an axis perpendicular to the motion direction of the cleaning robot **10**. In a case that the cleaning robot **10** needs to move to the base station body **20**, the rotating wheel **19** may cooperate with the guide block **25**, which facilitates the cleaning robot **10** to move to the base station body **20** more smoothly.

[0154] In an embodiment of the present disclosure, as shown in FIG. **23**, the guide part may include a guide wheel **26**. In addition to assisting the cleaning robot **10** in moving to the base station body **20**, the guide wheel **26** restricts the motion of the cleaning robot **10** in a vertical direction once the cleaning robot **10** stays at the base station body **20**. For example, in a case that the cleaning robot **10** stays at the base station body **20** for cleaning, the cleaning component **30** of the base station contacts the cleaning head **410** of the wet cleaning system **400** of the cleaning robot **10**, and applies a vertical upward thrust onto the cleaning robot **10**. The arrangement of the guide wheel **26** may partially or completely offset the vertical upward thrust, thereby preventing the cleaning robot **10** from moving upwards. Optionally, there may be at least two guide wheels **26**, and they are symmetrically distributed on the left and right sides of the base station.

[0155] The base station in an embodiment can realize the cleaning of the cleaning robot, the liquid filling of the liquid container of the cleaning robot, and the charging of the cleaning robot.

[0156] Other embodiments of the present disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the contents disclosed herein. The present application is intended to cover any variations, uses, or adaptations of the present disclosure, which are in accordance with the general principle of the present disclosure and include common general knowledge or conventional technical means in the art that are not disclosed in the present disclosure. The specification and embodiments are illustrative, and the real scope and spirit of the present disclosure is defined by the appended claims.

[0157] It should be understood that the present disclosure is not limited to the precise structures that have been described above and shown in the drawings, and various modifications and changes can be made without departing from the scope thereof. The scope of the present disclosure is limited only by the appended claims.

Claims

1. A base station for cleaning a cleaning system of a cleaning robot, comprising: a base station body, comprising a cleaning groove; a cleaning component, movably disposed on the base station body, and comprising a liquid outlet device, wherein cleaning liquid discharged by the liquid outlet device is configured to clean the cleaning system and enter the cleaning groove; and a driving part, connected with the base station body and configured to drive the cleaning component to move relative to the cleaning robot, wherein the cleaning component further comprises a cleaning part, and the cleaning part comprises a first cleaning part and a second cleaning part; the first cleaning part and the second cleaning part are separately disposed to be movable synchronously in a same direction.
2. The base station according to claim 1, wherein the cleaning component further comprises: a cleaning component holder, on which the liquid outlet device is disposed.
3. The base station according to claim 2, wherein the driving part is further connected with the cleaning component holder and configured to drive the cleaning component holder to move relative to the base station body.
4. The base station according to claim 2, wherein the base station further comprises: a liquid delivery channel, wherein one end of the liquid delivery channel is in communication with a liquid supply part, and the other end of the liquid delivery channel is in communication with the liquid outlet device, so that the liquid supply part supplies the cleaning liquid into the liquid outlet device via the liquid delivery channel; wherein, at least a part of the liquid delivery channel is movably disposed along with the cleaning component holder.

5. The base station according to claim 1, wherein a plurality of liquid outlets are disposed on the liquid outlet device at intervals.

6. The base station according to claim 2, wherein the cleaning part is disposed on the cleaning component holder parallel to the liquid outlet device.

7. The base station according to claim 6, wherein the first cleaning part and the second cleaning part are configured to remove debris from the cleaning system by interacting with the cleaning system.

8. The base station according to claim 7, wherein the first cleaning part comprises a cleaning roller configured to be rotatable while moving horizontally, thereby bringing debris out from a cleaning head of the cleaning system.

9. The base station according to claim 7, wherein the second cleaning part comprises a cleaning scraper configured to scrape the debris and waste liquid on the cleaning head.

10. The base station according to claim 7, wherein a liquid outlet is disposed on the liquid outlet device, wherein the liquid outlet is disposed to face at least one of the first cleaning part and the second cleaning part, and the cleaning liquid discharged from the liquid outlet is sprayed to at least one of the first cleaning part and the second cleaning part.

11. The base station according to claim 1, wherein a liquid outlet is disposed on the liquid outlet device, the liquid outlet is disposed to face the cleaning system, and the cleaning liquid discharged from the liquid outlet is sprayed to the cleaning system.

12. The base station according to claim 1, further including a liquid delivery channel, wherein one end of the liquid delivery channel is in communication with a liquid supply part, and the other end of the liquid delivery channel is in communication with the liquid outlet device, the liquid delivery channel is provided with a pump, and the cleaning liquid within the liquid supply part is delivered to the liquid outlet device under the action of the pump.

13. A cleaning robot system, comprising: a base station for cleaning a cleaning system of a cleaning robot; and a cleaning robot, wherein the base station comprises: a base station body, comprising a cleaning groove; a cleaning component, movably disposed on the base station body, and comprising a liquid outlet device, wherein cleaning liquid discharged by the liquid outlet device is configured to clean the cleaning system and enter the cleaning groove; and a driving part, connected with the base station body and configured to drive the cleaning component to move relative to the cleaning robot, wherein the cleaning component further comprises a cleaning part, and the cleaning part comprises a first cleaning part and a second cleaning part; the first cleaning part and the second cleaning part are separately disposed to be movable synchronously in a same direction.

14. A base station for cleaning a cleaning system of a cleaning robot, comprising: a base station body, comprising a cleaning groove; a cleaning component, disposed on the base station body approximately horizontally, and comprising a liquid outlet device and a cleaning part; and a driving part, connected with the base station body and configured to drive the cleaning component to move relative to the cleaning robot, wherein, cleaning liquid discharged by the liquid outlet device is configured to clean the cleaning system by the cleaning part and enter the cleaning groove, wherein the cleaning component further comprises a cleaning part, and the cleaning part comprises a first cleaning part and a second cleaning part; the first cleaning part and the second cleaning part are separately disposed to be movable synchronously in a same direction.

15. The base station according to claim 14, wherein the cleaning component is movably disposed on the base station body.

16. The base station according to claim 14, wherein the cleaning component further comprises: a cleaning component holder, on which the liquid outlet device and the cleaning part are disposed, wherein the liquid outlet device is parallel to the cleaning part.

17. The base station according to claim 14, wherein a plurality of liquid outlets is disposed on the liquid outlet device at intervals.

18. The base station according to claim 14, further comprising: a first pump, configured to pump the cleaning liquid into the cleaning groove; and a second pump, configured to pump out the cleaning liquid from the cleaning groove.

19. The base station according to claim 16, wherein the driving part is further connected with the cleaning component holder and configured to drive the cleaning component holder to move relative to the base station body.

20. The base station according to claim 14, further comprising: a liquid filling connector, disposed on the base station body, and configured to connect with a liquid filling port of a liquid container of the cleaning robot, to fill liquid into the liquid container of the cleaning robot.
